

Build Your First AI Agent

Last month, we learnt what agentic AI is and how it works. In this second article on agentic AI, we will learn how to develop and design an AI-driven multi-agent service.



The growth of agentic artificial intelligence (AI) is being fuelled by the complexity of the tasks that machines do. AI agents are becoming an essential part of our technological landscape, be it for self-driving cars or conversational robots. At the heart of agentic AI is the architecture that determines how agents sense, learn, decide, and act. The tools and frameworks used to construct such agents are manifold, each bringing the capability to work on specific problems. Navigating through these platforms is overwhelming but is crucial for designing effective, smart agents.

In essence, agentic AI refers to systems that can make choices and act on their own without the need for constant human interference. The most typical use of this kind of AI is found in autonomous agents, which do the work of moving around spaces, solving problems, and engaging with humans or other machines. To construct such agents one needs to have a firm grasp of both theoretical principles and working tools.

Developers use specialised frameworks that provide tools for creating and managing AI agents efficiently. Here are some of the most important agentic AI frameworks used today.

Microsoft AutoGen

AutoGen is an open source programming framework for building AI agents and facilitating cooperation among multiple agents to solve tasks. It is designed to handle multi-agent systems where different AI agents collaborate to solve complex problems. It enables AI agents to work independently or with human input, allowing for flexible and scalable AI solutions.

Key features:

- Supports multi-agent workflows and real-time collaboration.
- Integrates human-in-the-loop interactions for oversight and decision-making.
- Provides memory and context management for AI conversations.
- Includes error-handling mechanisms to ensure reliability.

Use case: A cloud management AI system where different agents work together to detect and fix server issues autonomously.